

Final Project

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Data Story and Manipulation

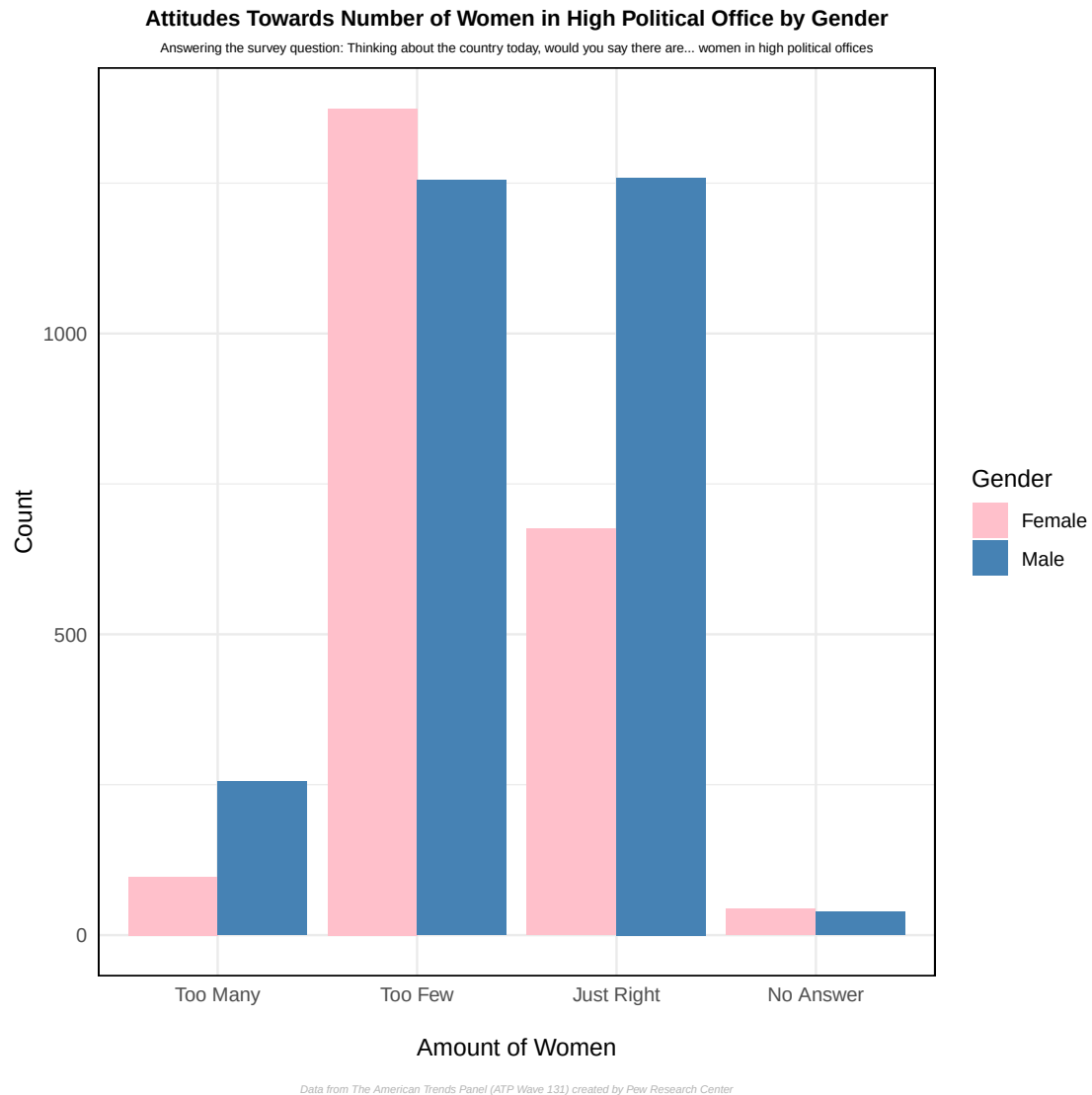
The Data description

```
1 #Data
2 leadership <- read_sav("~/Desktop/Trinity/DataViz_2026/Final_Project/ATP W131.
   sav")
3
4 #Manipulation
5 leadership_2 <- leadership %>%
6   select(
7     F_GENDER, #gender
8     F_EDUCAT2, #education level
9     F_AGECA1, #age category
10    F_USR_SELFID, #self identified area (1 = Urban, 2 = Suburban, 3 = Rural)
11    RESPECTW1_W131, #level of respect for female president
12    AMNTWMNBF1_W131, #level of feeling for amount of women in top executive
    business positions
13    AMNTWMNPF1_W131, #level of feeling for amount of women in high political
    offices
14  ) %>%
15  rename(
16    respect_female = RESPECTW1_W131,
17    amount_exect = AMNTWMNBF1_W131,
18    amount_polit = AMNTWMNPF1_W131
19  ) %>%
20  mutate(
21    F_GENDER = case_when(
22      F_GENDER == 1 ~ "Male",
23      F_GENDER == 2 ~ "Female",
24      TRUE ~ NA_character_ # everything else becomes NA
25    ),
26    F_GENDER = factor(F_GENDER)
27  ) %>%
28  filter(!is.na(F_GENDER))
```

Data Visualization

Visualization 1

```
1 #1
2 pdf("bar.pdf")
3 ggplot(leadership_2, aes(x = factor(amount_polit), fill = F_GENDER)) +
4   geom_bar(position = "dodge") +
5   scale_fill_manual(values = c("Male" = "steelblue", "Female" = "pink")) +
6   scale_x_discrete(
7     labels = c(
8       "1" = "Too Many",
9       "2" = "Too Few",
10      "3" = "Just Right",
11      "99" = "No Answer"
12    )
13  ) +
14  labs(
15    title = "Attitudes Towards Number of Women in High Political Office by Gender",
16    subtitle = "Answering the survey question: Thinking about the country today, would you say there are women in high political offices",
17    x = "\nAmount of Women",
18    y = "Count",
19    fill = "Gender",
20    caption = "\nData from The American Trends Panel (ATP Wave 131) created by Pew Research Center"
21  ) +
22  theme_minimal() +
23  theme(
24    plot.title = element_text(hjust = 0.5, face = "bold", size = 10),
25    plot.subtitle = element_text(hjust = 0.5, size = 6),
26    panel.border = element_rect(color = "black", fill = NA, size = 0.7),
27    panel.background = element_rect(fill = "white", color = NA),
28    strip.background = element_rect(fill = "white", color = "black", size = 0.7),
29    plot.caption = element_text(face = "italic", size = 5, color = "grey70", hjust = 0.5)
30  )
31 dev.off()
```



Visualization 2

```

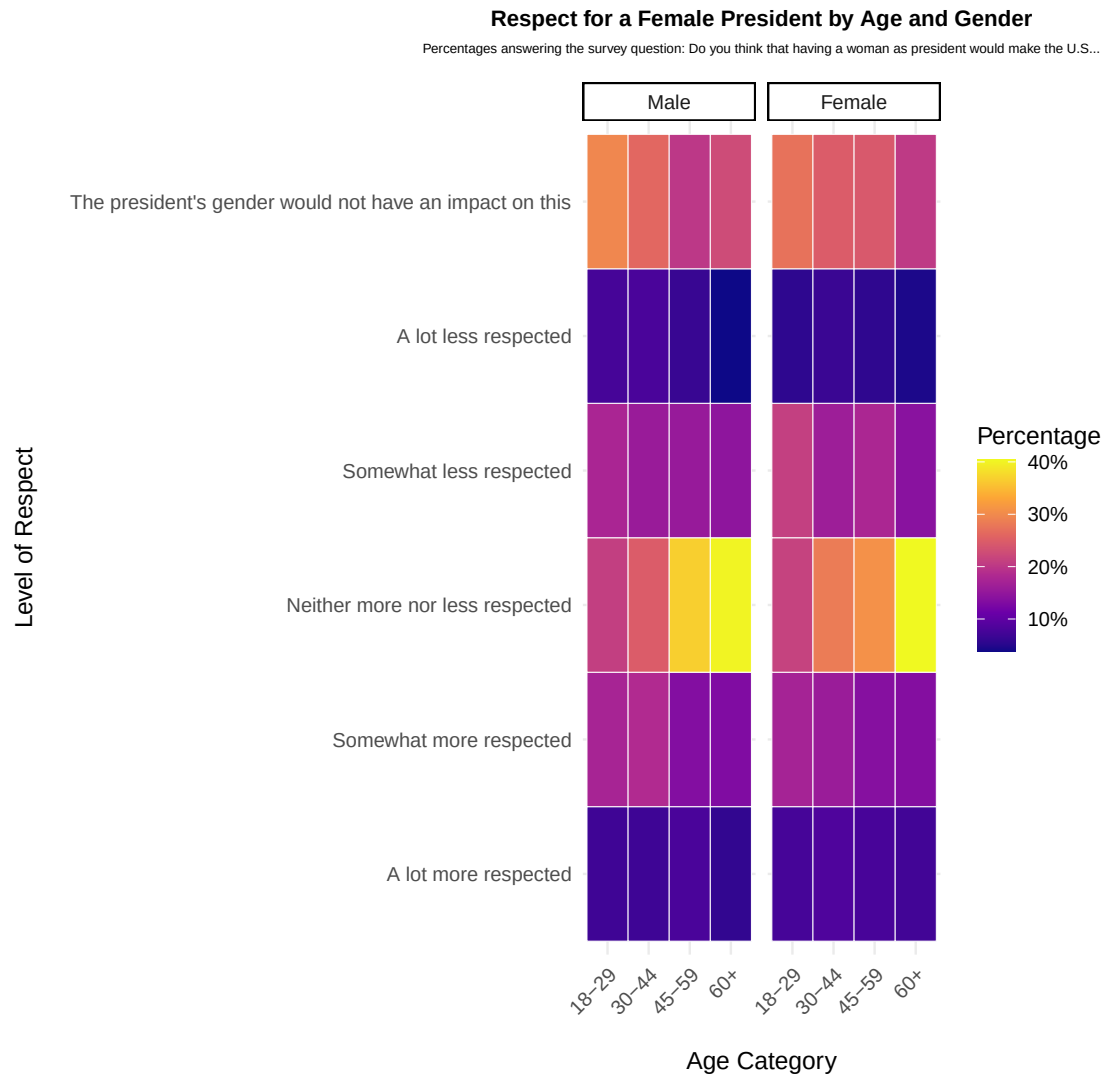
1 #2
2 #organizing the categories to fit the message
3 leadership_clean <- leadership_2 %>%
4   mutate(respect_female = as_factor(respect_female, levels = "labels")) %>%
5   filter(
6     F_AGECA1 %in% 1:4,
7     !is.na(respect_female),
8     respect_female != "Refused"
9   ) %>%
10  mutate(
11    F_AGECA1 = factor(F_AGECA1,
12                      levels = 1:4,
13                      labels = c("18-29", "30-44", "45-59", "60+")),
14    respect_female = factor(respect_female, ordered = TRUE),

```

```

15   F_GENDER = factor(F_GENDER, levels = c("Male", "Female"))
16   )
17
18   #calculating the percentages of each respect level based on age category and
    gender
19   respect_age_gender <- leadership_clean %>%
20     count(F_GENDER, F_AGECA, respect_female) %>%
21     group_by(F_GENDER, F_AGECA) %>%
22     mutate(pct = n / sum(n)) %>%
23     ungroup()
24
25   pdf("heat.pdf")
26   ggplot(respect_age_gender, aes(x = F_AGECA, y = respect_female, fill = pct))
    +
27     geom_tile(color = "white") +
28     facet_wrap(~ F_GENDER) +
29     scale_fill_viridis_c(labels = percent_format(), option = "C") +
30     labs(
31       title = "Respect for a Female President by Age and Gender",
32       subtitle = "Percentages answering the survey question: Do you think that
    having a woman as president would make the U.S...\n",
33       x = "\nAge Category",
34       y = "Level of Respect\n",
35       fill = "Percentage",
36       caption = "\nData from The American Trends Panel (ATP Wave 131) created
    by Pew Research Center"
37     ) +
38     theme_minimal() +
39     theme(
40       plot.title = element_text(hjust = 0.5, face = "bold", size = 10),
41       plot.subtitle = element_text(hjust = 0.5, size = 6),
42       strip.background = element_rect(fill = "white", color = "black", size =
    0.7),
43       axis.text.x = element_text(angle = 45, hjust = 1),
44       plot.caption = element_text(face = "italic", size = 5,
    color = "grey70", hjust = 0.5)
45     )
46   )
47   dev.off()

```



Data from The American Trends Panel (ATP Wave 131) created by Pew Research Center

Visualization 3

```

1 #renaming the factors for education
2 leadership_2 <- leadership_2 %>%
3   filter(F_EDUCCAT2 != 99) %>% # remove 99 first
4   mutate(
5     F_EDUCCAT2 = factor(
6       F_EDUCCAT2,
7       levels = sort(unique(F_EDUCCAT2)),
8       labels = c("Less than HS", "HS Grad",
9                 "Some College", "Associates", "Bachelors",
10                "Postgrad")
11     )
12   )
13 
```

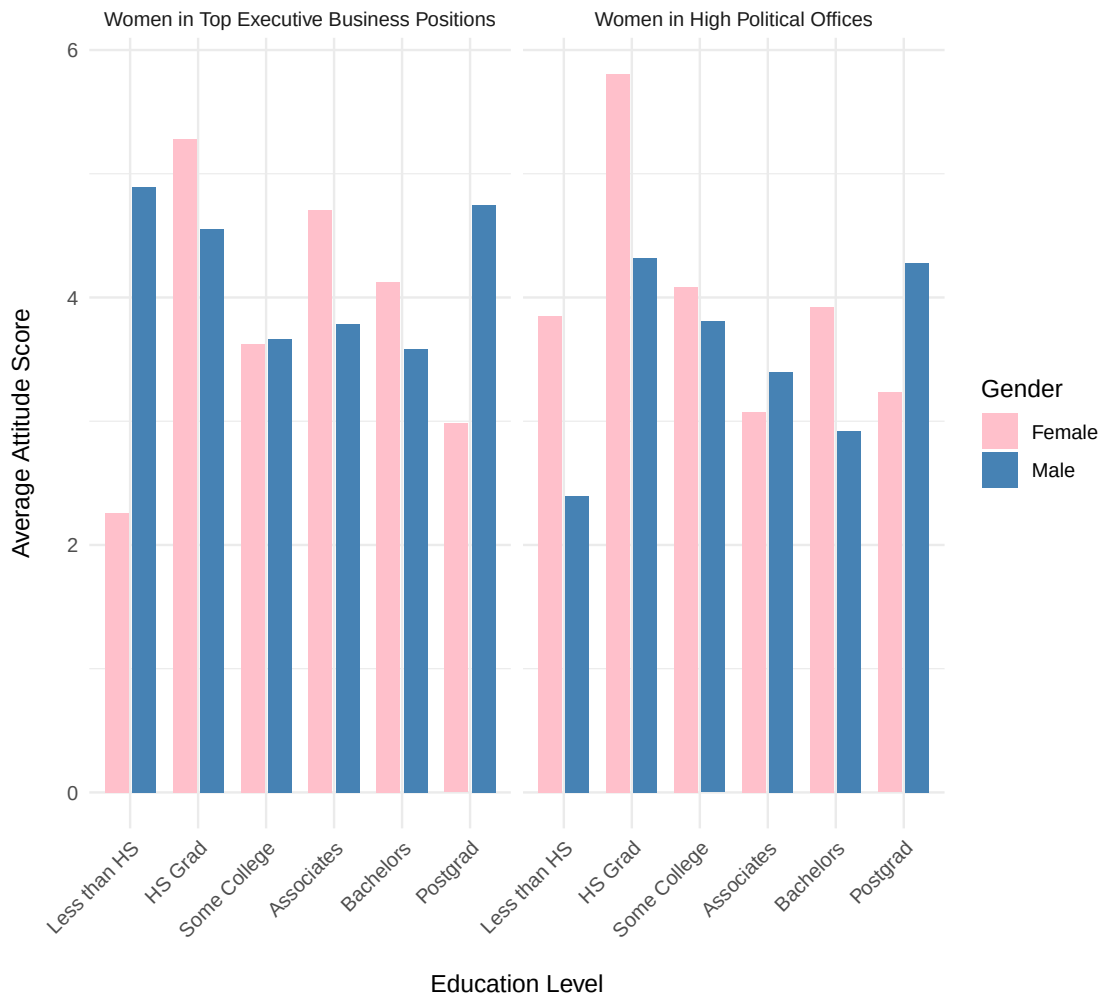
```

14 #pivoting the data to be long so we can look at all the education levels per
    gender
15 leadership_2$amount_exect
16 leadership_long <- leadership_2 %>%
17   pivot_longer(
18     cols = c(amount_exect, amount_polit),
19     names_to = "attitude_type",
20     values_to = "score"
21   )
22
23 pdf("bar_2.pdf")
24 ggplot(leadership_long,
25       aes(x = F_EDUCCAT2, y = score, fill = F_GENDER)) +
26   scale_fill_manual(values = c("Male" = "steelblue", "Female" = "pink")) +
27   stat_summary(fun = mean, geom = "bar",
28               position = position_dodge(width = 0.8), width = 0.7) +
29   facet_wrap(~attitude_type,
30             scales = "fixed",
31             labeller = as_labeller(c(amount_exect = "Women in Top Executive
    Business Positions",
32                                     amount_polit = "Women in High Political
    Offices")))) +
33   labs(
34     title = "Attitudes Toward Women in Leadership by Gender and Education",
35     subtitle = "The average values of opinions on the amount of women in these
    postions (1 = too many) per education category",
36     x = "\nEducation Level",
37     y = "Average Attitude Score\n",
38     fill = "Gender",
39     caption = "\nData from The American Trends Panel (ATP Wave 131) created by
    Pew Research Center"
40   ) +
41   theme_minimal() +
42   theme(
43     plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
44     plot.subtitle = element_text(hjust = 0.5, size = 8),
45     axis.text.x = element_text(angle = 45, hjust = 1),
46     plot.caption = element_text(face = "italic", size = 5,
47                                 color = "grey70", hjust = 0.5)
48   )
49 dev.off()

```

Attitudes Toward Women in Leadership by Gender and Education

The average values of opinions on the amount of women in these positions (1 = too many) per education category



Data from The American Trends Panel (ATP Wave 131) created by Pew Research Center